Classification of elements and Periodicity in Properties

① Dobereiner's law of triads : (Based on atomic weight)

→ Doberneiner formed groups of 3 elements (called triads) having similar properties where the atomic weight of middle element was an average of 1st & 3rd element.

→ Any type ~~bad~~ triads he formed



→ Any type ~~test~~ triads be formed having similar properties and near same atomic weight.

e.g. ^{7}Li, ^{23}Na, ^{39}K

e.g Ca Sr Ba



e.g. ^{7}Li, ^{23}Na, ^{39}K

e.g Ca Sr Ba

e.g Fe, Co, Ni

} Dobereiner's triads.

Drawback: Many elements could not fit into triads



Dobereiner : Many elements

② Newland's law of octaves : (Based on atomic mass)

→ Like every 8th musical note repeat itself, Newland arranged elements in increasing order of their atomic weight and suggest that every 8th element repeat its properties.



Li ~~Be~~ Be B C N O F

(Na) Mg Al Si P S Cl

→ Drawback : Did not think about the elements which were not discovered during that time.

After the discovery of noble gases (He, Ne, Ar),



9+ was found that every 9th element has similar properties rather than every 8th element.

(3) Lothar-Meyer curve : (Based on atomic weight)

→ Elements of similar properties occupied similar position in a curve plotted by him as atomic volume vs atomic weight.




atomic volume
Rb
K
Sr
Br
Na
Ca
Li
Be
F
Mg
Cl
Fe
Co
Ni
atomic weight

Subscribe



⟶ atomic weight

Features: ① Peak (or crests) were occupied by alkali metals.

② Troughs (or lowest points) are occupied by d- block metals

③ Alkaline earth metals occupy the m
of descending portion of curve



④ Halogens occupy the mid points of ascending portions of the curve.

Draw back : Lack of Practical utility and difficult to remember.



difficult

(4) Mendeleev's periodic table : (Based on atomic weight)

→ First organised classification in tabular form

→ Periodic table → 7 horizontal rows (called Periods)
→ 9 vertical columns (groups)

→



→ He arranged elements in increasing order of atomic weight and shown that ~~simp~~ elements of similar properties appear in the same group.

Merits of Mendeleev's Periodic table:

① First classification in orga




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Merits of Mendeleev's Periodic table:

1. First classification in organised tabular form.
2. Left vacant spaces in his periodic table for the elements that were not discovered during that time. He named these elements as eka-aluminium (later we called as




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for the elements that were not discovered during that time. He named these elements as eka-aluminium (later we called as gallium) and eka-silicon (now germanium)








elements as eka-aluminium (now gallium) and eka-silicon (now germanium)

Drawbacks: ① No clear position of hydrogen as it resembles in properties with alkali metals as well halogens.



Drawbacks: ① No clear position of hydrogen as it resembles in properties with alkali metals as well halogens.

② Dissimilar elements were placed in same groups. e.g. Cu, Ag, Au (coinage metals) were placed along with alkali metals.

→ Mn (Manganese) was placed ~~in~~ along with halogens (non-metals)



were placed along ...

→ Mn (Manganese) was placed ~~in~~ along with halogens (non-metals)

Mendeleev's periodic law : Properties (Physical and chemical) of elements are the periodic function of their atomic weights.



Moseley's periodic law :

→ Physical and chemical properties of elements are the periodic function of their atomic numbers.



→ Physical and chemical properties of elements are the periodic function of their atomic numbers.

→ Mosely bombarded highly energetic beam of electron on a metal studied the frequencies of x-rays emitted from metal. He found that fre








frequency of emitted x-rays are closely related with atomic no. of metal (& not with mass)

$$\sqrt{\nu} = a(Z - \sigma)$$



With Mass)

$$\sqrt{\nu} = a(Z - \sigma)$$ ⟵

where ν = frequency of emitted X-rays

Z = atomic No.

a = Proportionality constant

σ = Screening constant



with mass)

$$\sqrt{\nu} = a(Z - \sigma) \quad \text{---(1)}$$

where ν = frequency of emitted X-rays

Z = atomic No.

a = Proportionality constant

σ = Screening constant or shielding constant.



σ = Screening Constant

Constant.

→ Atomic number (Z) is the more fundamental property which decides the physical and chemical properties of elements.



properties of elements.

⇒ For understanding of students:

→ For electronic transition,

$\Delta E = E_{n_2} - E_{n_1} = h\nu$

$h\nu = \frac{-13.6Z^2}{n_2^2} + \frac{13.6Z^2}{n_1^2}$

$\Rightarrow \nu = \frac{13.6 \times Z^2}{h}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\nu \propto Z^2 \Rightarrow \sqrt{\nu} \propto Z$ ✓



Modern form of Periodic table (or Long Form) :

→ Developed by Bohr, Bury, Rang & Werner based on Moseley's periodic law.

→



→ Developed by Bohr, Bury, Rang & Werner based on Moseley's periodic law.

→ ~~Consists of~~ Consists of

→ 7 horizontal rows called periods

→ 18 vertical columns called groups or families.



→ 18 vertical columns called groups or families.

→ Period No. represents max. Principal quantum No (n) in Electronic conf^{n} of element.

4th period element → Max. P. Quantum No (n) = 4



Group No: 1 to 18 (as per IUPAC) =

IA, IIA -- etc (CAS system, chemical abstract system)

G No → / Period No ↓	1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9	10 VIII	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 zero
1								H										He
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg											Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe					Ga	Ge	As	Se	Br	Kr
5	Rb	Sr											In	Sn	Sb	Te	I	Xe
6	Cs	Ba											Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra																



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								VIII										
1								H										He
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg											Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	—	—	—	—	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba	La* 57	Hf 72	—	—	—	—	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra	Ac* 89	—	—	—	—	—	—	—	—	—						



Group No: 1 to 18 (as per IUPAC)

IA, IIA ... etc (CAS system, chemical abstract system)

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								VIII										
1 (2)								H										$_{2}He$ +8
2 (8)	Li	Be											$_{5}B$	C	N	O	F	$_{10}Ne$
3 (8)	Na	Mg											$_{13}Al$	Si	P	S	Cl	$_{18}Ar$ +8
4 (18)	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	$_{30}Zn$	$_{31}Ga$	Ge	As	Se	Br	$_{36}Kr$ +18
5 (18)	Rb	Sr	Y	Zr	—	—	—	—	Rh	Pd	Ag	$_{48}Cd$	$_{49}In$	Sn	Sb	Te	I	$_{54}Xe$ +18
6 (32)	Cs	Ba	La^{*} $_{57}$	Hf $_{72}$	—	—	—	—	Ir	Pt	Au	$_{80}Hg$	nTl	Pb	Bi	Po	At	$_{86}Rn$ +32
7	Fr	Ra	Ac^{*} $_{89}$	—	—	—	—	—	—	—	—	—	—	—				

incomplete (capacity : 32)



Group No: 1 to 18 (as per IUPAC) =

IA, IIA ... etc (CAS system, chemical abstract system)

G.No → / Period No ↓	1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9	10	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 zero
1 (2)	H																	He +8
2 (8)	Li	Be											B	C	N	O	F	Ne
3 (8)	Na	Mg											Al	Si	P	S	Cl	Ar +8
4 (18)	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr +18
5 (18)	Rb	Sr	Y	Zr	—	—	—	—	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe +18
6 (32)	Cs	Ba	La*	Hf	—	—	—	—	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	
7	Fr	Ra	Ac*	—	—	—	—	—	—	—	—	—	—	—				

s-Block | d-Block | p-Block | f-block

incomplete (Catalyst ...)

Redox Reaction
WHO'S TO BLAME?

7	Fr	Ra	Ac* 89	-	-	-	-	—	-	-	-	-	—	—			

incomplete (capacity : 32)

f-block

Lanthanide series (4f)	Ce 58	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Lu 71
Actinide series (5f)	Th 90	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	Lr 103



Lanthanide series (4f)	Ce 58	— — — — — — — — — — — — — —	Lu 71
Actinide series (5f)	Th 90	— — — — — — — — — — — — — —	Lr 103

Instead of Zn,Cd,Hg Coinage metals are Cu,Ag,Au

→ Zn, Cd, Hg are called Coinage metals due to ductile and malleable property

→
- → s-Block (outer E.C. : ~~$ns^{1 to 2}$~~ $ns^{1 to 2}$
- → p-block (outer EC : ns^2
-
-



→ Zn, Cd, Hg are called coinage metals due to ductile and malleable property

→
- → s-Block (outer E.C. : ~~$ns^{1 to 2}$~~ $ns^{1 to 2}$)
- → p-block (outer EC : $ns^2 np^{1 to 6}$)
- → d-block (outer EC : $(n-1)d^{1 to 10}\ ns^{0 to 2}$)
- → f-block (outer EC : $(n-2)f^{1 to 14}\ (n-1)d^{0 to 1}\ ns^2$)

} Imp

→



Redox Reaction

WHO'S TO BLAME?

→ f-block (outer EC : $(n-2)f^{1\text{to}14}\ (n-1)d^{0\text{to}1}\ ns^2$)

→ Nearly 78 % of elements are metals, remaining mostly non-metals and some metalloids

* Metalloids or semi metals
: Si, Ge, As, Sb, Te, Po, At
(They have properties of metals as well as non-metals



→ No of elements in a period is decided by possible orbitals as following (based on Aufbau's rule)

$ns, np, (n-1)d, (n-2)f, (n-3)g, (n-4)h$ ------

e.g 4th period, $\Rightarrow n = 4$

4s → 1
4p → 3
3d → 5
~~4f~~

9 orbitals × 2
= 18 elements



e.g 6th period, ⇒ n=6

6s, 6p, 5d, 4f, ~~3d~~

(1 + 3 + 5 + 7) = 16 orbitals

= 32 Elements

Q→ In future, if P.T. is expanded following Aufbau's & Pauli's rule, find no of elements that can be accomodated in 9th period.

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Q → In future, if P.T. is expanded following Aufbau's & Pauli's rule, find no of elements that can be accomodated in 9th period.

Soln: $n=9$, $9s^{\checkmark}, 9p^{\checkmark}, 8d^{\checkmark}, 7f^{\checkmark}, 6g^{\checkmark}, 5h$ ~~5h~~

→ $(1+3+5+7+9)$

= 25 orbitals

= 50 elements

1s
2p
3d
4f
5g
6h



Note: ① Periodic table can be divided into these three parts

- Representative elements
- Transition elements
- Inner transition elements



Note: ① Periodic table can be divided into these three parts

- Representative elements
  - s & p-block elements (Group NO: 1, 2, 13 to 18)
- Transition elements (d-block elements, all metals)
  - * Zn, Cd, Hg (Group NO = 12) are d-block elements
- Inner transition elements (f-block elements)
  - → all metals



s & p-block elements (Group NO: 1, 2, 13 to 18)

all metals)

* Zn, Cd, Hg (Group NO = 12) are d-block elements but are not regarded as transition elements.
* d-block elements which have d^1 to d^9, any configuration in either free form or in combined form are called Transition elements



② Typical elements : 3rd period elements (except noble gas) are called Typical elements. Other elements placed below these elements in same group have similar properties.



Na, Mg Al Si P S Cl

Typical elements

→ 2nd period elements (Li, Be, B, C, N, O, F) are not chosen as typical elements because they show different properties (exceptional behaviour) from rest of the elem



properties.

Na, Mg Al Si P S Cl

↑ Typical elements

→ 2nd period elements (Li, Be, B, C, N, O, F) are not chosen as typical elements because they show different properties (exceptional behaviour) from rest of the elements of same group.

It is due to (i) Small size
(ii) High Electronegativity



(Li, Be, B, C, N, O, F)

IUPAC nomenclature of elements having $Z > 100$:

→ These are temporary names given to new elements unless their name is finalized.



not chosen as typical elements because they show different properties (exceptional behaviour) from rest of the elements of same group.

It is due to
(i) Small size
(ii) High Electronegativity
(iii) Absence of d-~~sd~~ orbitals in valence shell.

in 2nd period elements
(Li, Be, B, C, N, O, F)



(Li, Be, B, C, N, O, F)

IUPAC nomenclature of elements having $Z > 100$:

↳ These are temporary names given to new elements unless their name is finalized.



digid	Latin word	Abbreviation
0	nil	n
1	un	u
2	bi	b
3	tri	t
4	quad	q
5	pent	p
6	hex	h
7	sept	s
8	oct	o
9	enn	e

→ A suffix 'ium' is applied in the name

eg ~~Z=105~~ Z = 105
(unnilpentium)

eg Z = 117
(ununseptium)









8	oct	o
9	enn	e

Predicting Group No, period No, block of an element :

→ Block of of an element is the orbital which receives last electron. (Exceptim : He)

→ Period No. is the principal quantum No. of outermost shell.

→ IUPAC group No (1 to 18) : For s-block, G. No = No. of ns electrons

For p-block, G. No = 10 + no. of np electrons + ns electrons.

For d-block, G. No = No of $(n-1)d\ e^-$ + no of ns electro

For f-block : G. N = 3 (fixed)



→ 2, 8, (8), 18, 18, 32, incomplete
(but can be max 32)

Q → why only 8 elements in 3rd period.

For n = 3; 3s, 3p, 3d

2 + 6 + 10 = 18 elements expected



Q → why only 8 elements in 3rd period.

For n = 3; 3s, 3p, 3d

2 + 6 + 10 = 18 elements expected

Ans: According to Aufbau's rule, before 3d is filled '4s' orbital has lower energy and is occupied.

In 3rd period: 3s, 3p

2 + 6 = 8 (elements)



$\Rightarrow$ ns, np, (n-1)d, (n-2)f, (n-3)g - - - - -

$\leq$ (Based on Aufbau's rule)

Effective nuclear charge (Z_{eff}):

(Screening effect or Shi



Effective nuclear charge (Z_{eff}):

(Screening effect or shielding effect)

In multielectron species, intervening electrons (inner electrons or electrons between nucleus



In multielectron species, ~~esse~~ intervening electrons (inner electrons or electrons between nucleus and valence shell) reduce the the force of attraction of nucleus on outer electrons. This reduction in force of attraction



nucleus and valence shell) reduce the the force of attraction of nucleus on outer electrons. This reduction in force of attraction by inner electrons is called screening effect or shielding effect.



Inner electrons shield the outer electrons from the attractive influence of nucleus.

Effective nuclear charge, $Z_{eff} = Z - \sigma$

where Z = Atomic Number

σ = shielding constant



where Z = Atomic Number

σ = shielding constant or screening constant.

→ For single $\bar{e}$ species:

$$E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

→ Considering shielding effect,

$$E_n = -13.6 \times \frac{Z_{eff}^2}{n^2} \text{ eV}$$

$\Rightarrow \sigma \uparrow$, $Z_{eff} \downarrow$



$\frac{}{n^2}$ eV

Factors affecting shielding constant (σ):

(1) Shielding is mainly caused by same shell electrons and inner electrons. outer electrons are not effecting in causing shielding.



and inner electrons. Outer electrons are not effecting in causing shielding.

(2) Closer is the electron to the nucleus, greater will be the shielding caused by it. (This is due to Penetration power)



it. (This is due to Penetration power)

For same shell (n),

$ns > np > nd > nf$

decreasing shielding caused
(per electron basis)

$3s > 3p > 3d$

better in shielding



Calculation of σ by slater's rule : (Not in IIT JEE syllabus)

(a) For (ns, np) electron :

(i) Write configuration as



(1s) (2s, 2p), (3s, 3p) (3d), (4s, 4p) (4d) (4f) ----

(ii) ~~Groups~~ Groups right to (ns, np) do not contribute in σ.

(iii) All the other electron contribute 0.35 towards σ. (However note that in case of 1s, contribution is 0.3)



(iii) All the other electron contribute 0.35 towards σ. (However note that in case of 1s, contribution is 0.3)

(iv) All the (n-1) shell $\bar{e}$ contribute 0.85 towards sigma.

(v) Rest all inner electrons contribute 1 towards σ.



towards sigma.

(v) Rest all inner electrons contribute 1 towards σ.

e.g "Na : $(1s^2)(2s^2 2p^6)(3s^1)$

σ on 3s electron:

$$\sigma = 0 \times 0.35 + 8 \times 0.85 + 2 \times 1$$

$$= 0 + 6.80 + 2 = 8.8$$



e.g "Na : $(1s^2)(2s^2 2p^6)(3s^1)$

σ on 3s electron:

$\sigma = 0 \times 0.35 + 8 \times 0.85 + 2 \times 1$

$= 0 + 6.80 + 2 = 8.8$

$(Z_{eff})_{on\ 3s} = 11 - 8.8 = 2.2$

σ on 2p electron:

$\sigma = 7 \times 0.35 + 2 \times 0.85$

$= 2.45 + 1.70 = 4.15$



$\sigma = 0 \times 0.35 + 8 \times 0.85 + 2 \times 1$

$= 0 + 6.80 + 2 = 8.8$

$(Z_{eff})_{on\ 3s} = 11 - 8.8 = \underline{2.2}$

σ on 2p electron :

$\sigma = 7 \times 0.35 + 2 \times 0.85$

$= 2.45 + 1.70 = 4.15$

$(Z_{eff})_{2p} = 11 - 4.15 = \underline{6.85}$



$(Z_{eff})_{2p} = 11 - 4.15 = 6.85$

(b) For nd or nf electron:

(i) Write conf^{n} as above

(ii) Electrons to the right of nd or nf do not contribute in σ.



(i) Write conf^{n} as above

(ii) Electrons to the right of nd or nf do not contribute in σ.

(iii) All the other electrons of nd or nf contribute 0.35 towards σ.

(iv) All inner electrons contribute 1 towards σ.



(iv) All inner electrons contribute 1 towards σ.

e.g. $_{30}Zn$: $(1s^2)$ $(2s^2, 2p^6)$ $(3s^2, 3p^6)$ $(3d^{10})$, $(4s^2)$ → X

$(\sigma)_{on\ 3d} = 9 \times 0.35 + 18 \times 1$

$= 3.15 + 18 = 21.15$

$(Z_{eff})_{3d} = 30 - 21.15 = \underline{8.85}$



$= 3.15 + 18 = 21.15$

$(Z_{eff})_{3d} = 30 - 21.15 = \underline{8.85}$

Note: $_2He : (1s^2)$

$\sigma = 1 \times 0.3 = 0.3$

$Z_{eff} = 2 - 0.3 = \underline{1.7}$



Periodicity in properties

(or Trends in properties in Periodic table)

(1)



(or Trends in properties in Periodic table)

(1) Atomic radius (or atomic size):

It is the distance between nucleus and outer most electrons in an atom.

⟶ Exact size of



most electrons in an atom.

→ Exact size of atom can not be determined as a single atom can't be isolated

(ii) Position of e^- can't be exactly determined (Heisenberg's Uncertainty principle)



principle)

→ Indirectly size (radius) of an atom of an element is determined indirectly using bond data (By x-ray diffraction or spectroscopy)



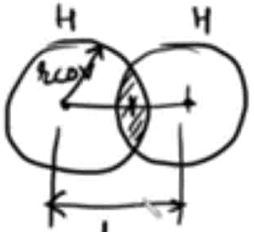
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spectroscopy)

(i) Covalent radius (r_{cov}):

It is one half of internuclear distance between [redacted] atoms in a homonuclear diatomic molecule (O_2, N_2, H_2, Cl_2 etc)





H_2

$r_{cov} = \frac{d}{2}$

→ Covalent radius is slightly smaller than the actual radius.

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→ In case of Heteronuclear molecule,



AB molecule

$$d_{AB} = r_A + r_B - 0.09\,|x_A - x_B|$$

Stevenson Formula.

d_{AB}, r_A, r_B → in Å

x_A, x_B → Electronegativity of x, y in



→ In case of Heteronuclear molecule,



AB molecule

$$d_{AB} = r_A + r_B - 0.09\,|x_A - x_B|$$

Stevenson Formula.

d_{AB}, r_A, r_B → in Å

x_A, x_B → Electronegativity of x, y in Pauling's scale



Pauling's scale.

→ Covalent radius mainly applies to non-metals as they form covalent bonds.

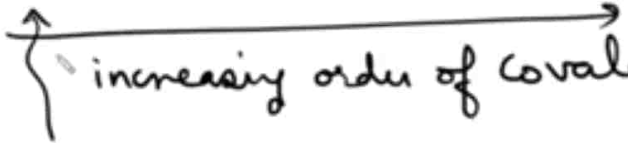
→ Between same pair of atoms,



as they form covalent bonds.

→ Between same pair of atoms,

TBCR < DBCR < SBCR



Triple bond
Covalent radius



Triple bond

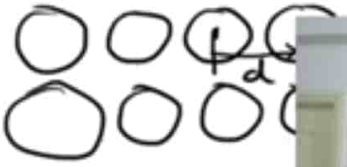
Covalent radius

→ SBCR is most accurate among these type of radii between a pair of atoms.

(ii) Metallic radius (r_{met}): It is one half of internuclear distance between two adjacent metal atoms in metallic lattice.

→ Mainly applies to metals.

$r_{met} = \frac{d}{2}$





→ Metallic radius is slightly larger than the actual size (radius) of metal atom.

(iii) Vanderwall's radius (r_{van})

→ It is also called non-bonding radius.



(iii) Vanderwall's radius (r_{van})

→ It is also called non-bonding radius.

→ It is mainly used for noble gases like He, Ne, Ar etc. which neither form a covalent bond nor exist as metallic ~~solid~~ lattice.



bond nor exist as metallic ~~solid~~ lattice.

→ It is one half of the internuclear distance of atoms in closest ~~proxmity~~ proximity in solid or liquid state.



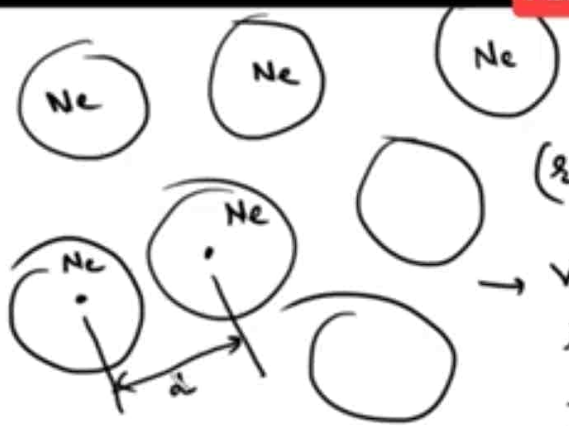
of atoms in closest ~~proximity~~ proximity in solid or liquid state.



$(r_{van})_{Ne} = \frac{d}{2}$

→ Vanderwaal's radius is much larger than the

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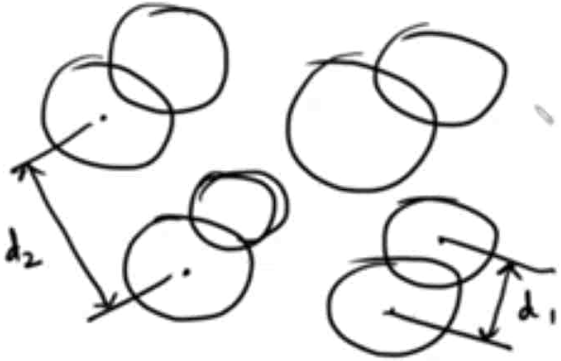


$(r_{van})_{Ne} = \frac{d}{2}$

→ Vanderwaal's radius is much larger than the actual radius (Highly erroneous)

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Ne(s)



$r_{cov} = \frac{d_1}{2}$

$r_{vdw} = \frac{d_2}{2}$

$H_2(s)$





$H_2(s)$

Note: For an element,

$r_{van} >> r_{met} > r_{cov}$

decreasing radii.



Factors affecting atomic radius :

① Z_{eff} : Z_{eff} ↑ , atomic size ↓

② No. of shells in atom (or Principal quantum No 'n')



② No. of shells in atom (or Principal quantum No. n)

$n \uparrow$, size $\uparrow$.

Trends of Z_{eff} in Periodic table (in s & p-block)



Trends of Z_{eff} in Periodic table (in s & p-block)

From L to R, Z_{eff} ↑

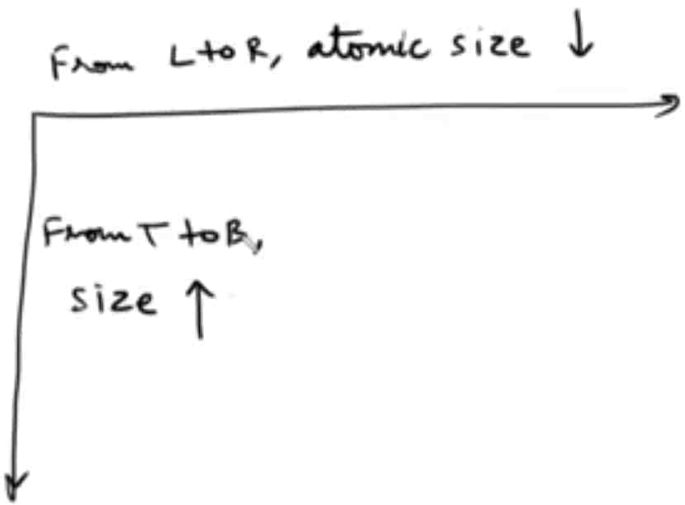
From T to B,

Z_{eff} remains nearly constant



Trends of atomic radius in P.T. (s & p-block):





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Exceptions:

① In 2nd, 3rd, 4th period, size of noble gas is found to be largest in the period. It is due to vanderwaal's radius used for noble gas.



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→ On uniform scale (i.e. all radii are vanderwaal's radii), there will not be any exception in radii of elements of



→ On uniform scale (i.e. Vanderwaal's radii), there will not be any exception in radii of elements of a period.

In

(2)



a period.

I.p (2) Atomic radius of Ga is slightly smaller than that of Al atom.

→ It is due to poor shielding caused by d' electrons that are filled in Ga.



(2) Atomic radius of Ga is slightly smaller than that of Al atom.

→ It is due to poor shielding caused by 'd' electrons that are filled in Ga.

$ns > np > nd > nf$

decreasing order of shielding caused.

→ 'd' & 'f' electrons are poor shielding electrons.



(2) Atomic radius of Ga is slightly smaller than that of Al atom.

→ It is due to poor shielding caused by 'd' electrons that are filled in Ga.

→ so Z_{eff} is high for Ga.

Al: $1s^2, 2s^2 2p^6, 3s^2 3p^1$

Ga: $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10} 4s^2 4p^1$

$ns > np > nd > nf$

decreasing order of shielding caused.

→ 'd' & 'f' electrons are poor shielding electrons.



③ Lanthanoid contraction : (or Lanthanoid Contraction)

3d Series:	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
4d Series:	Y	Zr	–	–	—	–	–	– – –	Ag	Cd
5d Series:	La*	Hf	–	–	– –	—	– – –		Ag	Hg – – –

$Sc < Y < La$ → size

Lanthanide contraction is observed here.

$Ti < \underbrace{Zr \cong Hf}_{\text{Lanthanide contraction.}}$

$Cu < Ag \cong Au$



→ when we move from 3d to 4d series (vertically down), atomic size increase as expected.

→ when we move from 4d to 5d series (vertically down), atomic size after La do not increase, rather slightly decreased or remains nearly



down), atomic size after La do not increase, rather slightly decreased or remains nearly same. This effect after La is called Lanthanoid contraction.

→ Reason for Lanthanoid contraction is the poor shielding caused by 14 e^- filled in f-sub



→ Reason for Lanthanoid contraction is the poor shielding caused by 14 e^- filled in f - subshell just befor H_f (hafnium). So Z_{eff} increaseses very sharply from La to H_f.



V. Imp

$Sc < Y < La$ (No exception)

$\xrightarrow{\text{atomic radii}}$

e.g. $Ti < Zr \cong Hf$

e.g. $Cu < Ag \cong Au$

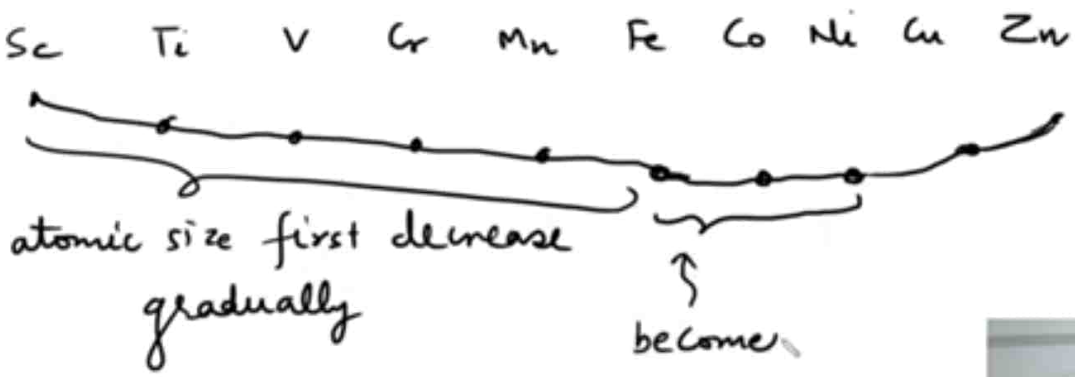


e.g. Cu < Ag ≅ Au

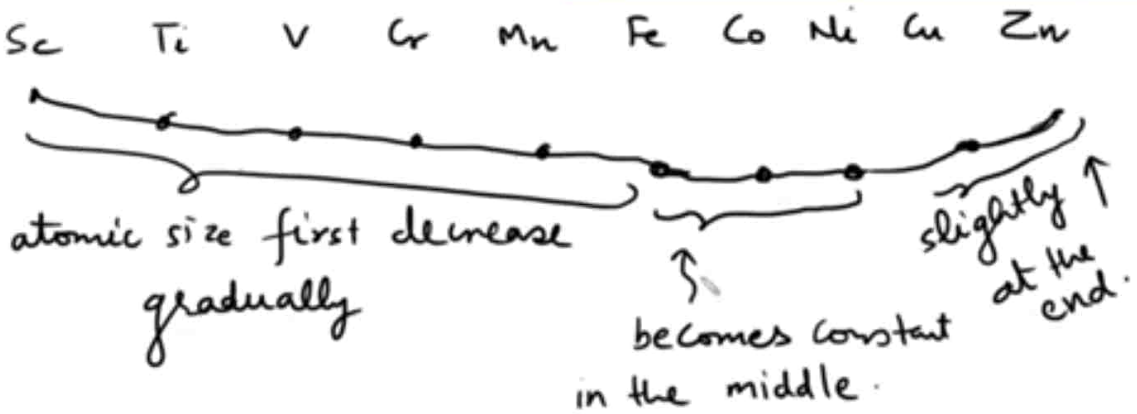
→ Effect of Lanthanoid contraction is also applicable in ionisation enthalpy and further into p-block-



Trends of atomic radii in 3d series elements :



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* Atomic radii of Cu & Zn are slightly larger (or increase) due to greater repulsion among electrons in fully filled d-subshell.



Ionic radius:

ion → cation
ion → anion

→ Size of cation is smaller than the size of its neutral atom.

→



of its neutral atom

→ Size of anion is larger than the size of its neutral atom.

* $F \longrightarrow F^-$

a pm $\quad \cong 2a$ pm



of its neutral atom.

* $F \longrightarrow F^-$
  a pm $\quad \cong 2a$ pm

* $Na \longrightarrow Na^+$ (one complete shell is removed)
  z pm $\quad$ nearly 70 to 80% size decrease occurs.

*



decrease occurs.

* $Mg > Mg^{+} >> Mg^{2+}$

one complete shell is removed.



one complete shell is removed.

V. Imp

Smallest atom : H

Largest atom : Cs

Smallest cation = H^+

Smallest anion = F^-



* $r_{H^-} \cong r_{Br^-}$

This means that H^- is enough large anion

eg $F \xrightarrow{a\ pm} F^- \cong 2a\ pm$

$\frac{e}{p} = \frac{9}{9}$ $\qquad$ $\frac{10}{9}$, % increase in $\frac{e}{p} = \frac{10}{9} - \frac{9}{9}$

$\frac{1}{9}$

$= 11\%$



$\frac{\quad}{1} \times 100$

$H \longrightarrow H^-$ (hydride ion) = 11 %.

$\frac{e}{p} = \frac{1}{1}$ $\qquad$ $\frac{e}{p} = \frac{2}{1}$, % increase in $\frac{e}{p}$ = 100 %.

* H^- is an unstable anion, therefore saline hydrides (or ionic hydrides) are



* H^- is an unstable anion, therefore saline hydrides (or ionic hydrides) are powerful reducing agent (e.g LiH, NaH, CaH_2, $LiAlH_4$, etc)

Powerful R.A.



Isoelectronic anions : ions having same no. of total electrons.

isoelectronic ions.

	N^{3-}	O^{2-}	F^{-}	Na^{+}	Mg^{2+}	Al^{3+}
$\frac{z}{e}$ =	$\frac{7}{10}$	$\frac{8}{10}$	$\frac{9}{10}$	$\frac{11}{10}$	$\frac{12}{10}$	$\frac{13}{10}$



→ Higher is $\frac{Z}{e}$ ratio, small is the size (radius) among isoelectronic ions

$$N^{3-} > O^{2-} > F^{-} > Na^{+} > Mg^{2+} > Al^{3+}$$

decreasing order of ionic radii



$N^{3-} > O^{2-} > F^{-} > Na^{+} > Mg^{2+} > Al^{3+}$

decreasing order of ionic radii

Note: IIT Adv.

$r_{Li^+} > r_{Mg^{2+}} > r_{Al^{3+}}$ $\Rightarrow$ (Remember)

0.76 Å 0.72 Å 0.53 Å

decreasing size

Q → Compare size of O^{2-}, S^{2-}, Cl^{-}, F^{-}

Soln: $S^{2-} > Cl^{-}$

$O^{2-} > F^{-}$

$S^{2-} > Cl^{-} > O^{2-} > F^{-}$

decreasing size



(2) Ionisation energy (IE):

It is the energy required to remove most loosely bound electron from a gaseous isolated atom to form gaseous cation.

→ Usually most loosely electron



loosely bound electron from a gaseous isolated atom to form gaseous cation.

→ Usually most loosely electron is the outermost electron which experiences least Z_{eff}.

$$M_{(g)} \longrightarrow M^{+}_{(g)} + e^{-}, \text{ Energy required} = IE_1(M_{(g)})$$



electron which experiences least Z_{eff}.

$M_{(g)} \longrightarrow M^{+}_{(g)} + e^{-}$, Energy required $= IE_1(M_{(g)})$

$M^{+}_{(g)} \longrightarrow M^{2+}_{(g)} + e^{-}$, $IE_2(M_{(g)}) = IE_1(M^{+}_{(g)})$

$M^{2+}_{(g)} \longrightarrow M^{3+}_{(g)} + e^{-}$, $IE_3(M_{(g)}) = IE_2(M^{+}) = IE_1(M^{2+}_{(g)})$

Note



$= IE_1 (M^{2+}_{(g)})$

Note:

For a given element, successive ionisation energy go on increasing.

For element M,

$IE_1 < IE_2 < IE_3 < - - - - -$



→ In thermodynamics convention, energy involved in a process or reaction are termed as enthalpy change (ΔH)

- → Endothermic rex or process
  - ↳ where energy is required (or absorbed), ΔH = +ve
- → Exothermic process,
  - ↳ energy liberates,

ΔH = -ve



Ionisation enthalpy (ΔH_{IE}): It is the enthalpy change when 1 mol of gaseous atoms of an element are converted into gaseous cation.

(Endothermic process)



(Endothermic process) change when 1 mol of gaseous atoms of an element are converted into gaseous cation.

$$M_{(g)} \longrightarrow M^{+}_{(g)} + e^{-}, \quad \Delta H_{IE_1}(n_{(g)})$$

1 mol

$$M^{+}_{(g)} \longrightarrow M^{2+}_{(g)} + e^{-}, \quad \Delta H_{IE_2}(M_{(g)}) = \Delta H_{IE_1}(M^{+}_{(g)})$$



Relation between IE and ΔH_{IE} : Imp

IE → is defined at zero Kelvin

ΔH_{IE} → is defined at any temp.

$$M_{(g)} \longrightarrow M^{+}_{(g)} + e^{-},$$

Taking $M_{(g)}$, $M^{+}_{(g)}$ and $\bar{e}$ to be monoatomic,



$M_{(g)} \longrightarrow M^+_{(g)} + e^-$, $(\Delta C_p)_R = \Sigma(C_p)_{prod} - \Sigma(C_p)_{react}$

Taking $M_{(g)}$, $M^+_{(g)}$ and e^- to be monoatomic,

Using Kirchoff's eqⁿ,

$$\frac{\Delta H_{T_2} - \Delta H_{T_1}}{T_2 - T_1} = (\Delta C_p)_R$$

Kirchoff's eqⁿ

$$\frac{\Delta H_{IE} - IE}{T - 0} = \frac{5}{2}R + \frac{5}{2}R$$



Taking $M_{(g)}$, $M^+_{(g)}$ and $\bar{e}$ to be monoatomic,

Using Kirchoff's eqⁿ,

$$\frac{\Delta H_{T_2} - \Delta H_{T_1}}{T_2 - T_1} = (\Delta C_p)_R$$

Kirchoff's eqⁿ

$$\frac{\Delta H_{IE} - IE}{T - 0} = \frac{5}{2}\left(\frac{5}{2}R + \frac{5}{2}R\right) - \frac{5}{2}R$$

$$\Rightarrow \boxed{\Delta H_{IE} = IE + \frac{5}{2}RT}$$

J/atom J/atom



Factors affecting IE :

1. Size ↑, IE ↓
2. Z_{eff} ↑, IE ↑
3. Special electronic configuration :



→ Half filled and fully filled conf^{n} are highly stable and therefore more energy is needed to remove electron from such conf^{n}.

Imp

(4) Penetration power:

For same

(4) Penetration power :

For same shell (n),

$$ns > np > nd > nf$$

decreasing penetration power

e.g 2s > 2p



$ns > np > nd > nf$

decreasing penetration power

e.g $2s > 2p$

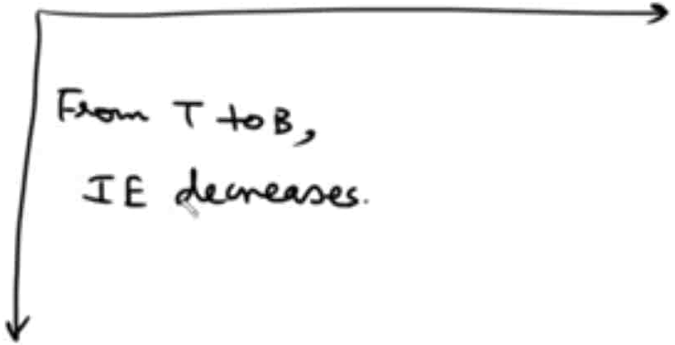
Removal of e^- from 2s requires more energy than that from 2p because penetration power of 2s is more.



Trends of IE in P.T. :

(general trends)

From L to R, IE increases





Exceptions:

① Li Be B C N O F Ne

Na Mg Al Si P S Cl Ar

$2s^2 2p^3$ $2s^2 2p^4$

$IE_{Be} > IE_B$

$IE_{Mg} > IE_{Al}$

$IE_N > IE_O$

$IE_p > IE_s$

half filled stable conf^n of Nitrogen.



Exceptions:

① Li | Be ($2s^2$) | B ($2s^2 2p^1$) | C | N ($2s^2 2p^3$) | O ($2s^2 2p^4$) | F | Ne

Na | Mg | Al | Si | P | S | Cl | Ar

$IE_{Be} > IE_B$

$IE_{Mg} > IE_{Al}$

higher amount of

$IE_N > IE_O$

$IE_p > IE_s$

half filled stable conf^n of Nitrogen.



$IE_{Mg} > IE_{Al}$

of Nitrogen.

higher amount of energy is required to remove e^- from 2s than 2p. (higher penetration power of 2s)



(2) IE of Ga is slightly greater than that of Al.

Ga > Al (5.98 ev/atom)

(IE = 6 ev/atom)

→ Reason: due to poor shielding caused by 10 electrons filled in 'd' subshell before Ga.



③ Lanthanide contraction :

After La,

$IE_{5d} > IE_{4d}$

$IE_{5d} > IE_{3d}$

e.g $IE_{Au} > IE_{Ag}$

$IE_{Au} > IE_{Cu}$

e.



$IE_{Au} > IE_{Cu}$

eg $IE_{Hg} > IE_{Cd}$

$IE_{Hg} > IE_{Zn}$

Reason: Very poor shielding caused by 14 electrons filled in '4f' subshell just after La (i.e. Ce to Lu, 58 to 71)



58 +1

Imp

→ The effect of Lanthanide contraction extends upto Pb (in p-block) in ionisation energy.

$IE_{Pb} > IE_{Sn}$

$IE_{Tl} > IE_{In}$

B	C
Al	Si
Ga	Ge
In	Sn
(Tl)	(Pb)



$IE_{Tl} > IE_{In}$

Ga	Ge
In	Sn
(Tl)	(Pb)

Applications of ionisation energy :

① Elements with low IE are electropositive in nature (said to have high metallic character)



nature (said to have high metallic character)

From L to R, Electropositive & nature or metallic nature ↓

From T to B,
metallic character ↑
or Electropositive character ↑

Li < Na < K < Rb < Cs

increasing metallic character



(2) Elements which have low IE, form basic oxides.

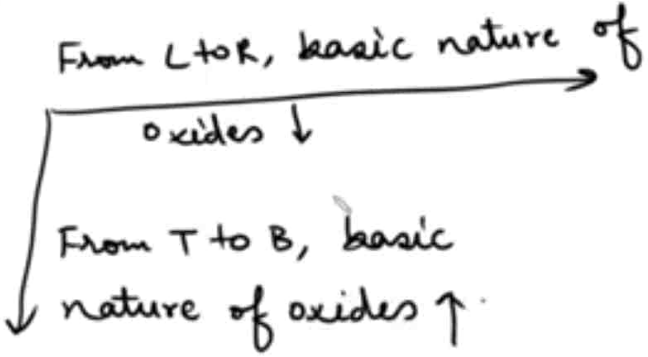
In general, alkali metal oxides are more basic than alkaline earth metal oxides.



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oxides.

In general, alkali metal oxides are more basic than alkaline earth metal oxides.





(3) Since metals easily lose electron (i.e. undergo oxidation), hence metals act as powerful reducing agent.

$$M_{(s)} \longrightarrow M^{+}_{(aq)} + e^{-}$$



reducing agent.

$$M_{(s)} \longrightarrow M^{+}_{(aq)} + e^{-}$$

→ On the basis of IE, Cs is expected to be the most powerful R.A.. However the fact is that in solution phase, Li is most powerful reducing metal (or element)



fact is that in solid ... most powerful reducing metal (or element)

→ Higher reducing power of Li is due to very high hydration energy (released) of Li^+ (small sized cation)



of Li^+ (small sized cation)

(4) Difference in successive ionisation energy helps in predicting stable oxidation state of an element (metals).

→ In general, a high difference in energy successive ionisation

an element (metals).

→ In general, a high difference in energy
successive ionisation energies indicate
greater stability of lower o.s.
(oxidation state)

→ A low difference in successive IEs
indicate greater stability of higher o.s.
(or valency)



Q → Element 'x',

$IE_1 = 5.7$ eV

$IE_2 = 36.2$ eV

Ans: Valency of element x is 1

Q → Metal 'M', $IE = 480$ KJ/mol

$IE_2 = 905$ KJ/mol

$IE_3 = 3850$ KJ/mol



Q → Metal 'M', IE_1 = 480 KJ/mol

IE_2 = 905 KJ/mol

IE_3 = 3850 KJ/mol

Find valency of 'M'

Sol^n: Ans: valency : 2 Ans

(eg Ca, Mg, Sr, etc
or Zn, Cd etc)



Q → Compare IE_2 of B, C, N, O

Soln:

B	C	N	O
$2s^2, 2p^1$	$2s^2, 2p^2$	$2s^2 2p^3$	$2s^2 2p^4$
$2s^2$	$2s^2 2p^1$	$2s^2 2p^2$	$2s^2 2p^3$
B^+	C^+	N^+	O^+

$C < B < N < O$

Increasing IE_2

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③ Electron affinity (EA):

It is the energy released when an electron is added to an isolated gaseous atom to form gaseous anion.



is added to an isolated gaseous atom to form gaseous anion.

$$X_{(g)} + e^- \longrightarrow X^-_{(g)}, \text{ Energy released}$$

= Electron affinity (EA_1) of (X)



$X_{(g)} + e^- \longrightarrow X_{(g)}$, Energy released

= Electron affinity

$(EA_1)_{(X)}$

* In most elements, EA is +ve i.e. energy is released when e^- is added.

However, in few cases, EA is -ve when energy is absorbed on adding e^-.



Electron gain enthalpy (ΔH_{eg}):

It is the enthalpy change when 1 mol of gaseous atoms of an element is converted into gaseous anion.



$$\underset{1\,mol}{X_{(g)}} + e^- \longrightarrow \underset{1\,mol}{X^-_{(g)}},\ \Delta H_{eg_1}$$

ΔH_{eg_1} is -ve for most elements as energy is released on adding $\bar{e}$.



However, in some elements ΔH_{eg} (or $\Delta H_{eg,1}$) is +ve in few cases where energy is absorbed.

Imp

→ In general, Electron gain enthalpy is the -ve of electron affinity in same energy units.



energy units.

$$X_{(g)} + e^- \longrightarrow X^-_{(g)},$$

$$\boxed{\Delta H_{eg} = -EA - \frac{5}{2}RT}$$

EA → defined at 0 'K'

ΔH_{eg} → defined at any temp.



ΔH_{eg} → defined at any temp.

→ Note that in comparison purpose or for comparing trends in PT, we use both the terms interchangeably.

$(\Delta H_{eg} = -EA)$



for comparing trends in PT,

we use both the terms interchangeably.

$(\Delta H_{eg} = -EA)$

e.g oxygen atom,

EA → +ve

ΔH_{eg} → -ve



$\Delta H_{eg} \rightarrow -ve$

V. Imp

Note :

$O_{(g)} + e^- \rightarrow \cancel{O^-_{(g)}}$, $EA_1\ (O_{(g)}) =$ +ve (energy is released)

$\cancel{O^-_{(g)}} + e^- \rightarrow O^{2-}_{(g)}$, $EA_2\ (O_{(g)}) =$ -ve (always) (energy is absorbed heavily)

$O_{(g)} + 2e^- \rightarrow O^{2-}$, $EA_1 + EA_2 =$ -ve (always)

---

or $\Delta H_{eg_1} + \Delta H_{eg_2} =$ +ve (



OR

$X_{(g)} + e^- \rightarrow X^-_{(g)}$, EA_1 = +ve in most cases

$X^-_{(g)} + e^- \rightarrow X^{2-}_{(g)}$, EA_2 = −ve (always) (heavily −ve)

$X_{(g)} + 2e^- \rightarrow X^{2-}_{(g)}$, $EA_1 + EA_2$ = −ve (always)

---

or $\Delta H_{eg_1} + \Delta H_{eg_2}$



$X_{(g)} + 2e^- \longrightarrow X^{2-}_{(g)}$, $EA_1 + EA_2 = -ve$ (always)

or $\Delta H_{eg_1} + \Delta H_{eg_2} = +ve$ (always)

(energy is absorbed, always)

(always endothermic)



→ P.E of attraction is −ve

P.E of repulsion is +ve.

If nucleus attraction for newly added e^- is dominant over repulsion by existing e^-, then energy is released

i.e EA = +ve
& ΔH_{eg} = −ve



ie EA = +ve
& ΔH_{eg} = -ve

Factors affecting EA :

① Size ↑, EA ↓

② Z_{eff} ↑, EA ↑



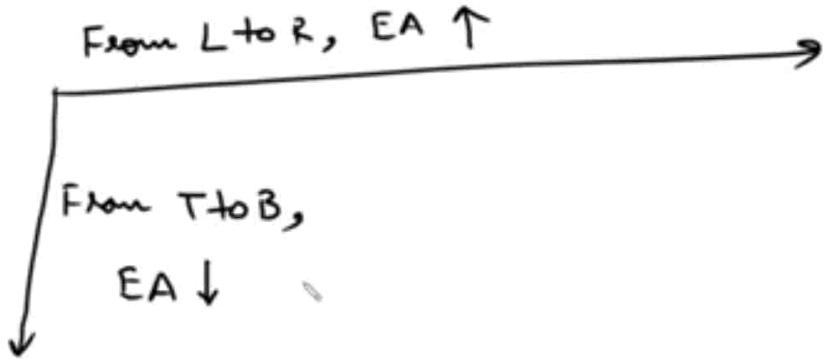
(3) Half filled and fully filled configuration are highly stable and have little tendency to add e^-. In other words, EA is mostly -ve (i.e. energy is absorbed) in such conf^n.



-ve (i.e. energy is absorbed) in such conf^{n}.

General Trends of EA in P.T.

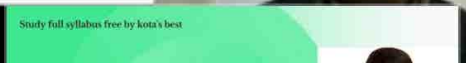



From L to R, EA ↑
From T to B,
EA ↓



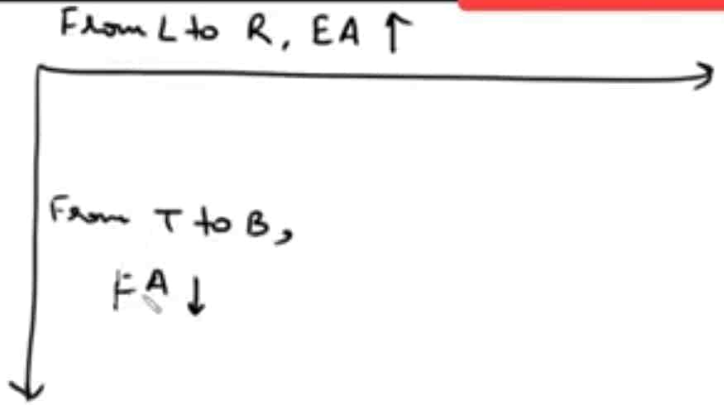

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Trends of EA in P.T.:





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↓

→ Trends of EA are less regular than that of ionisation energy.

Exceptions of EA :



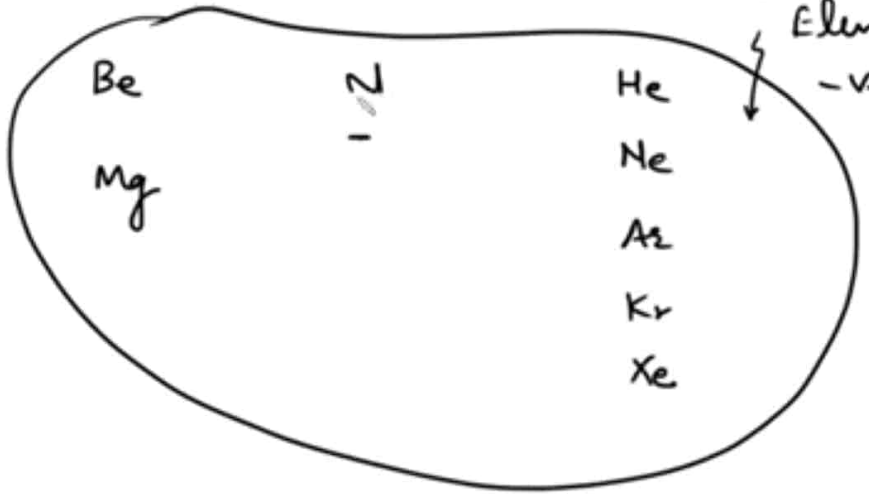
ionisation energy.

Exceptions of EA :

(1) Li [Be] B C [N] O F [Ne]



⇒ -ve EA is for the following notable elements



~~Eleven~~ Elements with -ve EA or +ve ΔH_{eg}



(or EA = -ve)

Xe

due to fully filled stable conf^n

due to half filled stable conf^n.

due to highly stable fully filled conf^n,

ΔH_{eg} = +ve for noble gases.



→ only 'N' in nitcogens has –ve EA.
(or +ve ΔH_{eg})

V. Imp

(2) EA of 3rd period 'p' block elements are greater than the corresponding 2nd period p-block elements.



p-block elements.

B C N O F ← 2nd period p-block

Al Si P S Cl ← 3rd period p-block.

Cl > F
S > O
P > N
Si > C
Al > B

} trends of EA.



P > N
Si > C
Al > B

EA.

* Highest EA ⟶ chlorine 'Cl'.

⟶ It is due to exceptionally small size of 2nd period p-block elements



→ It is due to exceptionally small size of 2nd period p-block elements. When electron is added, a very heavy repulsion is created by existing valence e^- of 2nd – period p-block elements due to high electron density (again due to very small size).



→ It is due to exceptionally small size of 2nd period p-block elements. When electron is added, a very heavy repulsion is created by existing valance e^- of 2nd – period p-block elements due to high electron density (again due to very small size).

So lesser energy is released than expected for 2nd – period p-block elements.



(8) ⇒ 'O' in chalcogens has smallest EA value.

S > Se > Te > Po > O

decreasing EA values

Cl > F > Br > I





(14) ⇒ 'O' in chalcogens has smallest EA value.

S > Se > Te > Po > O

decreasing EA values

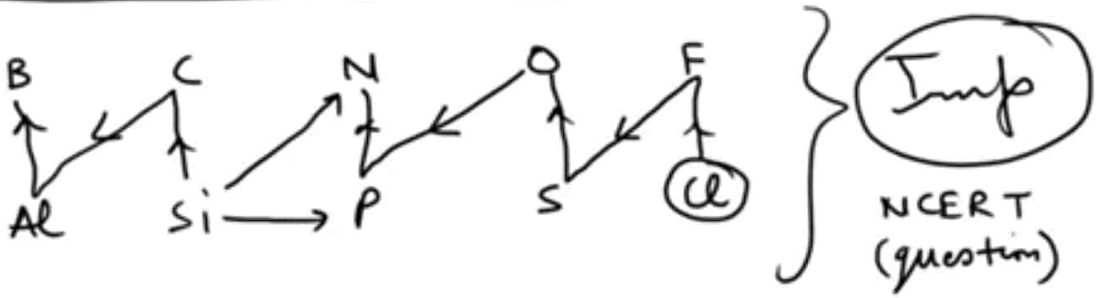
Cl > F > Br > I

decreasing EA values

V. Imp



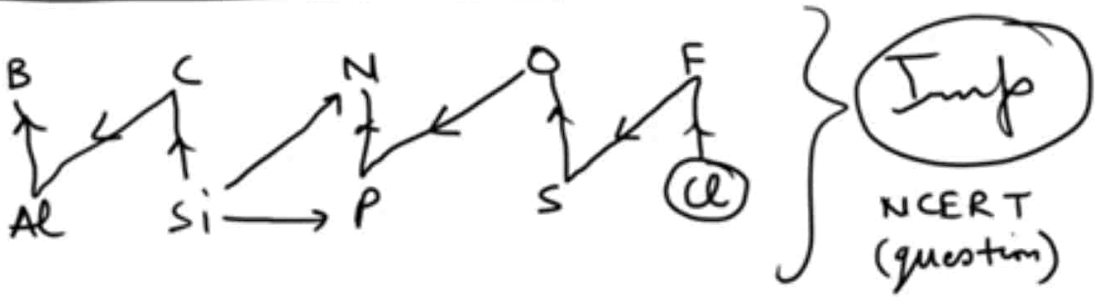
⇒ For EA Comparison



N < P < O < S < F < Cl

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⇒ For EA comparison



N < P < O < S < F < Cl

increasing EA

B < Al < C < Si



B < Al < C < Si

→ increasing E.A.

V. Imp

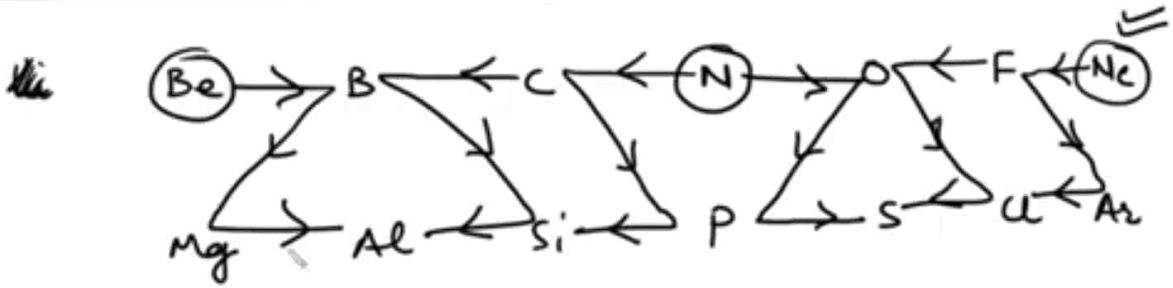
⇒ For Ionisation energy :



V Imp

increasing E.A.

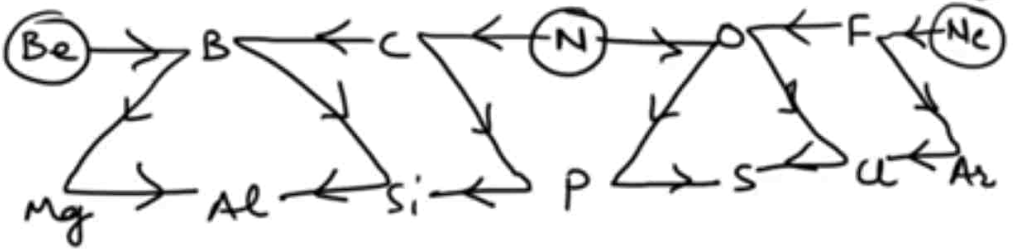
⇒ For Ionisation energy :



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for Ionisation energy :

Ie



Ne > F > Ar > Cl

decreasing IE_1

F > O > Cl > S

decreasing IE_1

N > O > P > S

decreasing IE_1



decreasing IE_1

⇒ The smallest value of IE of an element (Cs) is even greater than the highest value of EA of an element (Cl).



Applications of EA : ① It is used in calculation of electronegativity of an element (using Mulliken's scale)

② Generally non-metals (halogens, chalcogens) have high EA values.



element (using Mulliken's scale)

(2) Generally non-metals (halogens, chalcogens) have high EA values.

→ High EA of an element represents high tendency to gain electron and therefore such elements act as good oxidi



high tendency to gain electron ...

fore such elements act as good oxidizing agent.

On this basis, Cl_2 is expected to be the most powerful oxidizing element in P.T.. However, note that F_2 is most powerful oxidizing element in P.T..



the most powerful oxidizing element in P.T..

However, note that F_2 is most powerful oxidizing element in P.T.

→ Highest oxidizing power of F_2 is due to mainly high hydration energy of small F^- ion and low bond dissociation energy of F_2

$Cl_2 > Br_2 > (F_2) > I_2$ → decreasing B.E. ord



(4) Electronegativity (x)

↑ chi

→ It is defined as the ~~abo~~ ability of an atom to pull shared pair of electron towards itself in a molecule.



towards itself in a molecule.

* Electric Electronegativity is not the property of an isolated ~~be~~ atom. It is the property of an atom in a covalent bond.



* Property of an isolated atom are ionisation energy and electron affinity.

* Electronegativity is not a measurable quantity.



* Electronegativity is not a measurable quantity.



↑ Polar covalent bond

Covalent molecule (Polar)

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(Polar)

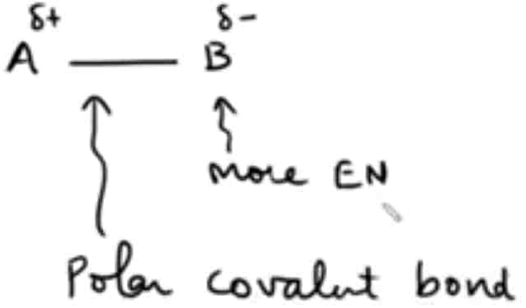
→ It determined in various scales

① Pauling scale of EN:



→ It determined in various scales

① Pauling scale of EN:



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$$X_A - X_B = 0.208 \sqrt{BE_{A-B} - \sqrt{BE_{A-A} \times BE_{B-B}}}$$

BE_{A-B}, BE_{A-A} and BE_{B-B} are in kCal/mol

$X_A X_B \rightarrow$ ~~Paulting~~ Pauling EN of A & B.



→ Originally Pauling assigned EN of H atom to be 2.1 and determined EN of other elements. (However, NCERT mentions that reference chosen by Pauling was Fluorine ('F' has 4.0)



Pauling values of EN,

F > O > N ≅ Cl > Br > C ≅ S ≅ I > H ≅ P > B > Li > Cs

4.0 3.5 3.0 3.0 2.8 2.5 2.5 2.5 2.1 2.1 2.0 1.0 0.7

Memorise (Useful)



4.0 3.5 3.0 3.0 2.8 2.5 2.5 2.5 2.1 2.1

Memorise (Useful)

(2) Mullikan Scale: According to Mullikan EN of an element is the average of IE and EA in ev. $\Rightarrow$ $EN_{Mullikan} = \frac{IE + EA}{2}$

$$EN_{Pauling} = \frac{EN_{Mullikan's\ scale}}{2.8}$$



Note: EN of an element may be different in different molecules.

Factors affecting EN



Factors affecting EN :

CORRECTION:LOWER IS THE SIZE, GREATER IS THE ELECTRONEGATIVITY

① Size ↑, EN ↑

② Zeff ↑, EN ↑



1. Size

CORRECTION:LOWER IS THE SIZE, GREATER IS THE ELECTRONEGATIVITY

2. $Z_{eff} \uparrow$, EN $\uparrow$

3. Oxidation Num



(3) Oxidation Number :

→ Higher is the +ve oxidation state, higher will be the EN.

eg $Fe < Fe^{2+} < Fe^{3+}$



→ higher is the +ve oxidation state, higher will be the EN.

eg $Fe < Fe^{2+} < Fe^{3+}$

eg $O > O^{2-}$

$F > F^{-}$

decreasing EN.



(4) State of hybridisation :

% s character in ~~hybrid~~ hybridised atom ↑,

EN of atom ↑

(This is due to greater penetration power of ns > np)



(This is due to greater penetration power of $ns > np$)

C_2H_6	C_2H_4	C_2H_2
sp^3	sp^2	sp
% s = 25 %	33.3 %	50 %



sp^3 sp^2 sp

% S = 25 % 33.3 % 50 %

$C_2H_6 < C_2H_4 < C_2H_2$

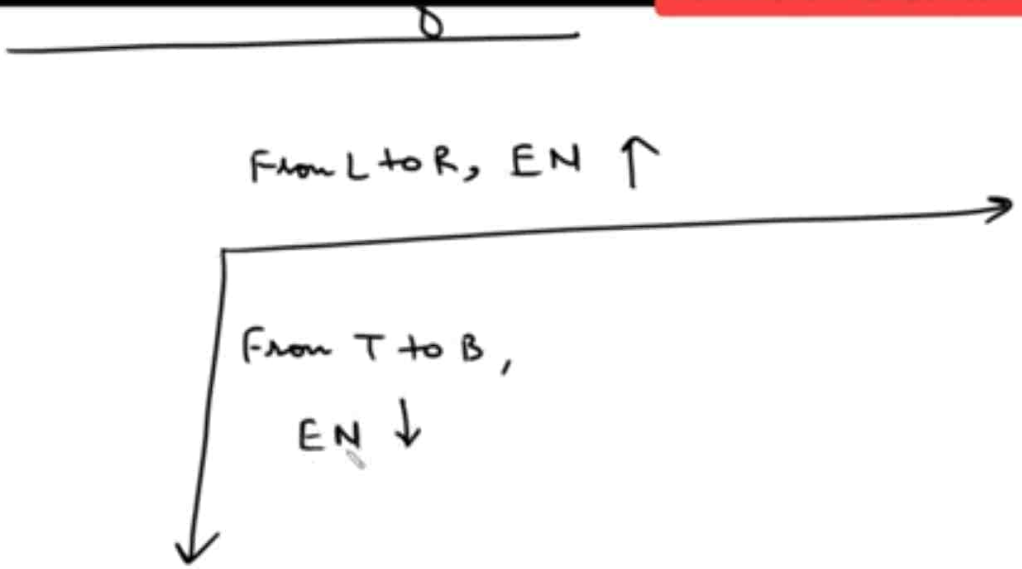
→ increasing EN of 'C' atom.



increasing EN of 'C' atom.

General Trends of EN




From L to R, EN ↑
From T to B,
EN ↓


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4.0 3.0 2.8 2.5

F > Cl > Br > I > At



3.5 2.8

O > S > Se > Te > Po



Li > Na > K ≥ Rb > Cs

1.0 0.9 0.8 0.8 0.7



Application of EN :

① % ionic character in a polar covalent bond



① % ionic character in a polar covalent bond :

$\overset{\delta+}{A} - \overset{\delta-}{B}$

↑ ↑ greater EN

Polar covalent bond



A — B

↑ ↖ greater EN

Polar covalent bond

Δ(EN) ↑ , % ionic character ↑



According to Hanny & Smith,

% ionic character in a bond

$$= 16\,|X_A - X_B| + 3.5\,(X_A - X_B)^2$$



$$= 16\,|X_A - X_B| + 3.5\,(X_A - X_B)^2$$

(According to Hanny & Smith,

If $\Delta(EN) = 2.1$, % ionic characte = 50 %.

$\Rightarrow$ If $\Delta(EN) > 2.1$, bond is said to be predominantly ionic bond.

$\Rightarrow$



⇒ If $\Delta(EN) < 2.1$, bond is predominantly Covalent.

→ More appropriately, According to Pauling,

If $\Delta(EN) = 1.7$, then 50 % ionic character

If $\Delta(EN) > 1.7$, bond is predominantly ionic



If $\Delta(EN) > 1.7$, bond is predominantly ionic

If $\Delta(EN) < 1.7$, bond is predominantly covalent.

e.g. HCl, $\Delta(EN) = 3.0 - 2.1 = 0.9$
(Polar covalent bond)

e.g. NaCl, $\Delta(EN) =$



e.g. HCl, $\Delta(EN) = 3.0 - 2.1 = 0.9$

(Polar covalent bond)

e.g. NaCl, $\Delta(EN) = 3.0 - 0.9$

$= 2.1 > 1.7$

(NaCl has ionic bonds)



$= 2.1 > 1.7$

(NaCl has ionic bonds)

(2) Higher is EN difference, in general, covalent bond will be stronger and shorter.



will be stronger and shorter.

$$H-F > H-Cl > H-Br > H-I$$

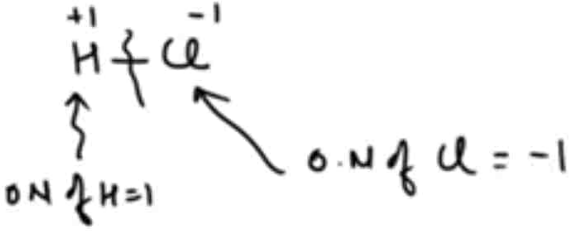
decreasing bond energy

$$H-F < H-Cl < H-Br < H-I$$

increasing Bond length



(3) Determination of oxidation Number :



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O.N of H = +1

O.N of Cl = −1

e.g CO_2 ⇒ $\overset{-2}{O} \overset{+2}{C} \overset{-2}{O}$

O.N of 'C' in CO_2 = +4

e.g SF_2 ⇒

$\overset{-1}{F} \overset{+1}{\ddot{S}} \overset{-1}{F}$

O.N of 'S' = +2



(4) Acidic or basic nature of X–OH molecule :

- $X\!-\!O\!-\!H \xrightarrow{H_2O(l)} X^+_{(aq)} + OH^-_{(aq)}$ (base)
  (when $EN_X < EN_H$)
  e.g. NaOH, KOH etc.



$X-O-H \xrightarrow{H_2O(l)} XO^-_{(aq)} + H^+_{(aq)}$ (acid)

[when $EN_X > EN_H$]

e.g ClOH, BrOH etc.

ClOH ← hypochlorous acid

BrOH ← hypobromous acid.



hypochlorous acid

→ More molar bond (in general) dissociates more easily in water due to very high di-electric constant of water (≅ 79)



(5) Elements with high electronegativity have high electronegative character.

→ 'F' is the most electronegative element in P.T.



F > O > N ≅ Cl > Br > S ≅ C ≅ I > H > B > Li > Na > Cs

4.0 3.5 3.0 ~~3.0~~ 5.0 2.8 2.5 2.5 2.5 2.1 2.0 1.0 0.9 0.7



→ Periodicity in chemical properties depend on electronic configuration. Elements with similar electronic conf^{n} (particularly outer E.C.) have similar chemical behaviour.

→ Periodicity in valence also depends on



→ Periodicity in valence also depends on electronic configuration.

Valence is defined as no of outer most electrons and/or ∞ (∞ - no of outer most electrons) for representative elements (s & p-block or 1, 2, 13-18 group)



Group No: 1, 2, 13, 14, 15, 16, 17, 18

No of valence e^- (outermost e^-) = 1, 2, 3, 4, 5, 6, 7, 8

Valence = 1, 2, 3, 4, 5, 6, 7, 8



→ Now a days a more appropriate term, called oxidation is more commonly used.

Valency
- Covalency (in covalent compounds) — No. of covalent bonds formed.
- Electrovalency (in ionic bonds) — No. of e^- lost or gained by an element
  ⇒ ionic compounds.



e.g $[Al(H_2O)_6]^{3+}$

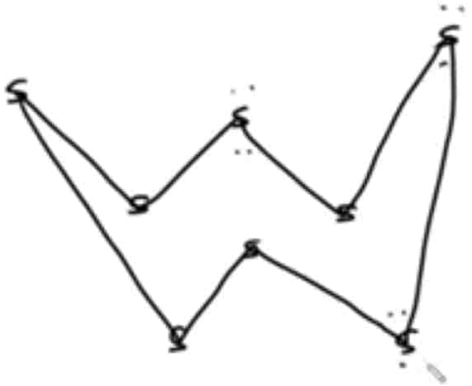
Covalency of Al = 6

Oxidation No of Al = +3

e.g Sulphur (S_8)



e.g Sulphur (S_8)



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Covalency of sulphur = 2
O.N. of sulphur = 0

C.g $BaCl_2$

Electrovalency of Ba = 2
Electrovalency of Cl = 1



Trends of acidic and basic nature of oxides :

→ Acidic oxides are anhydrides of oxyacids.

e.g $H_2SO_4 \xrightarrow[-H_2O]{} SO_3$

↑

an anhydride of H_2SO_4.



→ Acidic oxides are anhydrides of oxyacids.

e.g $H_2SO_4 \xrightarrow{-H_2O} SO_3$

↑ an anhydride of H_2SO_4.

e.g $2HNO_3 \xrightarrow{-H_2O} N_2O_5$

↑ anhydride of HNO_3.

→ Basic oxides are anhydrides of metal hydroxide

e.g $2NaOH \xrightarrow{-H_2O} Na_2O$ | $Ba(OH)_2 \xrightarrow{-H_2O} BaO$



Types of oxides :

① **Acidic oxides** : Generally oxides of non-metals which react with bases

→ If soluble, form acidic solution.

(acidic solution turn blue litmus red)



litmus red)

e.g. SO_3, SO_2, SiO_2, CO_2, N_2O_5, N_2O_3, Cl_2O_7 etc

(2) Basic oxides : → Generally the oxides of metals

→ React with acids

→ form basic solution in



litmus red)

e.g. SO_3, SO_2, SiO_2, CO_2, N_2O_5, N_2O_3, Cl_2O_7 etc

(2) Basic oxides : → Generally the oxides of metals

→ React with acids

→ form basic solution in water if soluble



e.g Na_2O, BaO, CaO, FeO, Fe_2O_3 etc.

(3) Amphoteric oxides : → These oxides react with acids as well as base.



base.

e.g. BeO, Al_2O_3, ZnO, Cr_2O_3, V_2O_5,

PbO, PbO_2, SnO, SnO_2, Sb_2O_3, As_2O_3 etc.

e.g. Al_2O_3 $\xrightarrow{HCl(aq)}$ $AlCl_3(aq) + H_2O(l)$

Al_2O_3 $\xrightarrow{NaOH(aq)}$ $NaAlO_2$



PbO, PbO_2, SnO, SnO_2, Sb_2O_3, As_2O_3 etc.

e.g. Al_2O_3 $\xrightarrow{HCl(aq)}$ $AlCl_3(aq) + H_2O(l)$

Al_2O_3 $\xrightarrow{NaOH(aq)}$ $NaAlO_2(aq) + H_2O$

Sod. metaaluminate



e.g. V_2O_5 $\xrightarrow{HCl(aq)}$ $VO_2^+(aq) + Cl^-(aq) + H_2O$ (oxy cation)

V_2O_5 $\xrightarrow{NaOH(aq)}$ $VO_4^{3-}(aq)$

e.g. ZnO $\xrightarrow{HCl(aq)}$

ZnO $\xrightarrow{NaOH(aq)}$



$VO_{3(aq)}$

e.g. ZnO

$ZnO + HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_2O_{(l)}$

$ZnO + NaOH_{(aq)} \rightarrow Na_2ZnO_{2(aq)} + H_2O_{(l)}$

Sod. Zincate

or $Na_2[Zn(OH)_4]$



$\rightarrow Na_2ZnO_2(aq) + H_2O(l)$

Sod. Zincate

or $Na_2[Zn(OH)_4]$

(4) Neutral oxide : → Neutral towards acid as well as base i.e do not react with acids & bases.



read with acids & bases.

N_2O, NO, CO

V. Imp

(5) Mixed



e.g ClO_2 is an anhydride of

(+4)

$HClO_2$ & $HClO_3$



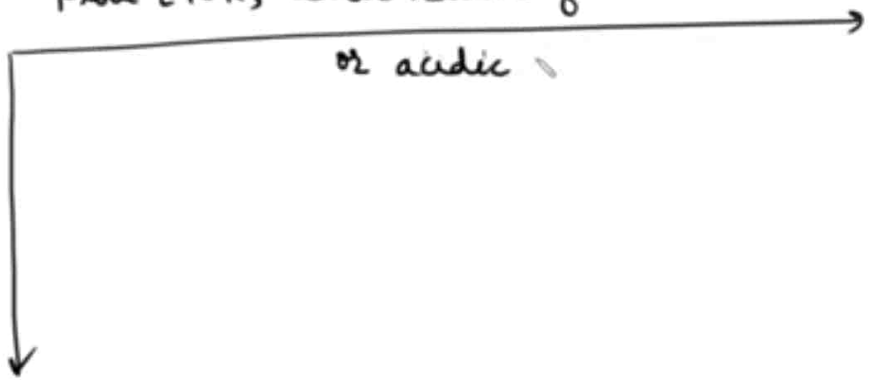
Trends of nature of oxides in P.T.



→ Metallic character ↑, basic nature ↑.

From L to R, basic nature of oxide ↓

or acidic





From L to R, basic nature of oxide ↓

or acidic nature ↑

From T to B,

basic nature ↑,

and acidic nature ↓



basic nature ↓,
and acidic nature ↑

Basic amphoteric

$Na_2O < MgO < Al_2O_3 < SiO_2 < P_4O_{10} < SO_3 < Cl_2O_7$

increasing acidic nature

weakly acidic

strongly acidic



$Li_2O > BeO > B_2O_3$

→ decreasing basic nature

$Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$

→



decreasing basic nature

$Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$



increasing basic nature.

$BeO < MgO < - - - - - < BaO$



increasing basic nature.



$BeO < MgO < - - - - - < BaO$

→ increasing basic nature

$Cl_2O > Br_2O > I_2O$

→ decreasing acidic nature

$ClOH > BrOH > IOH$

→ decreasing acidic nature

$HClO_3 > HBrO_3 > HIO_3$

→ decreasing acidic nature



Note: when an element forms oxyacids in two or more o.s (oxidation state), In general

o.s ↑, acidic nature ↑

$H_2SO_4 > H_2SO_3$

$HNO_3 > HNO_2$



$H_2SO_4 > H_2SO_3$

$HNO_3 > HNO_2$

(+5) (+3)

More acidic

or

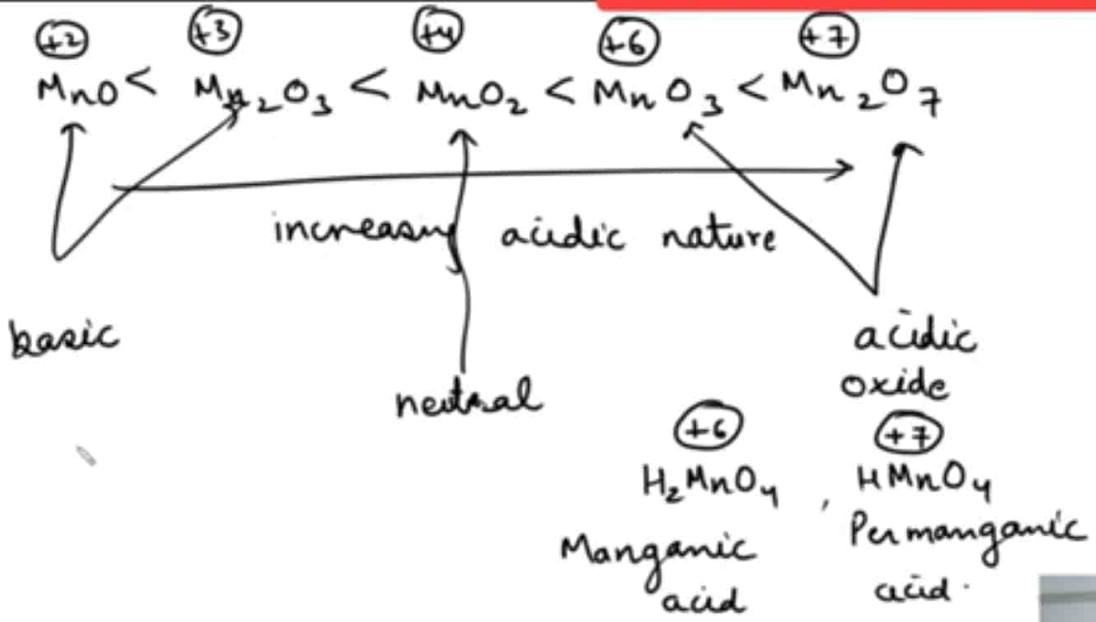
$SO_3 > SO_2$

$H_2O_5 > N_2O_3$

(+5) (+3)

More acidic




+2
+3
+4
+6
+7
MnO < Mn2O3 < MnO2 < MnO3 < Mn2O7
increasing acidic nature
basic
neutral
acidic oxide
+6
H2MnO4
Manganic acid
+7
HMnO4
Permanganic acid


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(JEE

Exception:

Imp

$H_3PO_3 \geq H_3PO_2 > H_3PO_4$

(+3) (+1) (+5)

pKa values: 1.1 1.2 2.1

$\longrightarrow$ pKa ↓, acidic strength ↑



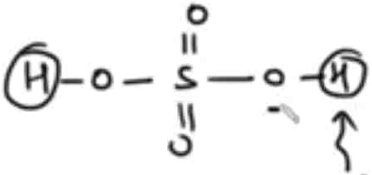
Oxiacids : Where 'H' is attached to 'O' atom and 'O' is again attached to some other electronegative element.

e.g. H_3PO_4, H_2SO_4, HNO_3, CH_3COOH, etc.



e.g. H_3PO_4, H_2SO_4, HNO_3, CH_3COOH, etc.

e.g.



(oxy acid) acidic Hydrogen

(dibasic acid or diprotic acid)



or diprotic acid)

Hydracids or (covalent hydrides):

↳ where 'H' is attached to some electronegative atom.

e.g HF, HCl, HBr, HI, H_2O, H_2S, H_2Se, $\ddot{N}H_3$, PH_3 etc.



Imp

From L to R, acidic nature of covalent hydrides ↑

From T to B, acidic nature of covalent hydride ↑



e.g * $CH_4 < NH_3 < H_2O < HF$ (In a period, EN is determining factor, EN ↑, acidic nature ↑)

$SiH_4 < PH_3 < H_2S < HCl$

increasing acidic strength



e.g

$HF < HCl < HBr < HI$ →

$H_2O < H_2S < H_2Se < H_2Te$ →

$NH_3 < PH_3 < AsH_3 < SbH_3$ →

increasing acidic nature

In a group, acidic nature is determined by size of atom and B.D.E (bond dissociation energy)

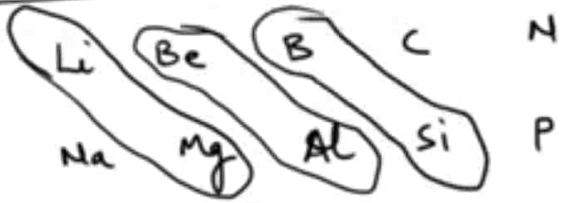


Size ↑, BDE ↓,

acidic nature ↑.



Diagonal relationship :



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These diagonal pair show many similar properties in their compounds & called Diagonal relationship.

→ Diagonal relationship is due to nearly same $\frac{\text{charge}}{\text{size}}$ values of pair elements.



These diagonal [illegible]

many similar properties in their compounds.

& called Diagonal relationship.

→ Diagonal relationship is due to nearly same $\frac{\text{charge}}{\text{size}}$ values of pair elements (ions).

e.g Both BeO & Al_2O_3 are amphoteric

e.g Li_2CO_3 & $MgCO_3$ on thermal decomposition give CO_2 gas.

E.g Both Be_2C and Al_4C_3 give CH_4 gas on hydroly

